Collision Risk Prediction

A PROJECT REPORT

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ABSTRACT

Autonomous vehicles (AVs) are transforming modern transportation, particularly in mission-critical applications like Autonomous Ambulance Systems (AAS), where minimizing response time and maximizing safety are vital. This project focuses on developing a Level 4 intelligent emergency response framework by predicting collision risks through data analytics and machine learning.

The study utilizes the KITTI dataset to extract high-dimensional features, including object density and proximity to surrounding obstacles. These features are processed through an Exploratory Data Analysis (EDA) phase to identify patterns and outliers before being used to train a Random Forest classification model. The model is designed to categorize driving scenarios as high-risk or low-risk, effectively handling the non-linear relationships inherent in complex urban traffic data.

The system's performance is evaluated using key metrics, including accuracy, precision, recall, and F1-score, ensuring the model meets the safety-critical standards required for emergency medical services. The results demonstrate that the proposed model can reliably identify hazardous situations based on traffic density and distance, highlighting the power of data-driven approaches in enhancing the Decision Layer of autonomous ambulances. This work provides a foundation for future integration with real-time V2X communication and deep learning models to further optimize emergency routing.

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I – Problem Definition

Objective

The objective of this project is to develop a data-driven model to predict collision risk in autonomous ambulance systems using features derived from real-world driving datasets. The goal is to enhance safety and improve decision-making.

Key Questions

- Can collision risk be predicted using object density and distance information?
- Does a decrease in distance to nearby objects increase the likelihood of a collision?
- Does higher traffic density (number of surrounding objects) lead to higher risk?
- Can machine learning models effectively classify safe and high-risk driving scenarios?

II – Literature Review

2.1 Introduction

Autonomous vehicles use artificial intelligence and sensor data to navigate without human intervention. In emergency situations, autonomous ambulances must travel quickly while ensuring safety in dynamic traffic conditions. One important task is collision risk prediction, which identifies dangerous situations.

This project focuses on using data from the KITTI dataset to build a machine learning model that predicts collision risk. Recent research in autonomous vehicle (AV) data analytics focuses on improving safety through collision risk prediction, object detection, and trajectory forecasting. These approaches combine sensor data with machine learning models to identify potential crash scenarios in real time.

2.2 Foundations and Latest Research

A. Perception and Object Detection

Modern autonomous vehicle systems rely heavily on perception models to detect objects such as vehicles, pedestrians, and road elements. Traditional CNN-based approaches perform object detection, while advanced Transformer-based models improve contextual understanding of scenes. These methods enable accurate identification of surrounding objects, which is essential for collision risk estimation.

B. Predictive Modelling and Risk Estimation

Recent research focuses on predicting vehicle and pedestrian behavior using machine learning and deep learning models. Techniques such as Random Forest is widely used to estimate collision probability based on features like distance, speed, and traffic density. These models help in identifying high-risk scenarios in real time.

C. Sensor Fusion and Multimodal Analytics

Autonomous systems integrate data from multiple sensors such as LiDAR, radar, and cameras. Sensor fusion techniques combine this data to improve accuracy and reliability. Advanced models use multimodal learning to enhance environmental understanding and reduce uncertainty in predictions.

D. Reliability and Uncertainty Estimation

Safety-critical systems require reliable predictions. Recent studies focus on uncertainty estimation in deep learning models to ensure robustness. These approaches help identify uncertain predictions and improve decision-making in complex driving environments.

Recent Research Summary

Year	Authors	Research Area	Problem	Methodology	Dataset	Key Findings
2024	Sun et al.	Perception	Need for large-scale AV data	Multi-sensor perception models	Waymo	Improved object detection accuracy
2023	Caesar et al.	Scene Understanding	Complex urban environments	Multimodal data analysis	nuScenes	Better environment modeling
2023	Shao et al.	Sensor Fusion	Unsafe decision-making	Transformer-based fusion	CARLA	Improved safety and context awareness
2022	Chitta et al.	Fusion Models	Lack of global context	Transformer-based learning	CARLA	Enhanced driving performance
2016	Redmon et al.	Object Detection	Real-time detection challenge	YOLO model	VOC/COCO	Fast and accurate detection

2.3 Benchmarking Existing Methods

Method	Application	Strength	Limitation
YOLO / CNN	Object detection	Real-time detection	High computational cost
LSTM / RNN	Trajectory prediction	Captures temporal patterns	Complex training
Random Forest	Collision risk prediction	Simple & interpretable	No time-series modeling
Sensor Fusion Models	Multi-sensor analytics	High accuracy	Requires heavy data

2.4 Methodology

The methodology focuses on transforming raw dataset information into meaningful features and applying for classification.

Stage 1: Data Collection and Representation

The first stage involves collecting data from the KITTI dataset, which contains labelled information about objects in driving scenes.

Stage 2: Data Preprocessing and Feature Engineering

In this stage, raw data is cleaned and transformed into useful features.

Stage 3: Model Development (Classification)

A machine learning model is developed to classify collision risk.

Stage 4: Prediction and Decision Support

The trained model is used to predict collision risk for new scenarios.

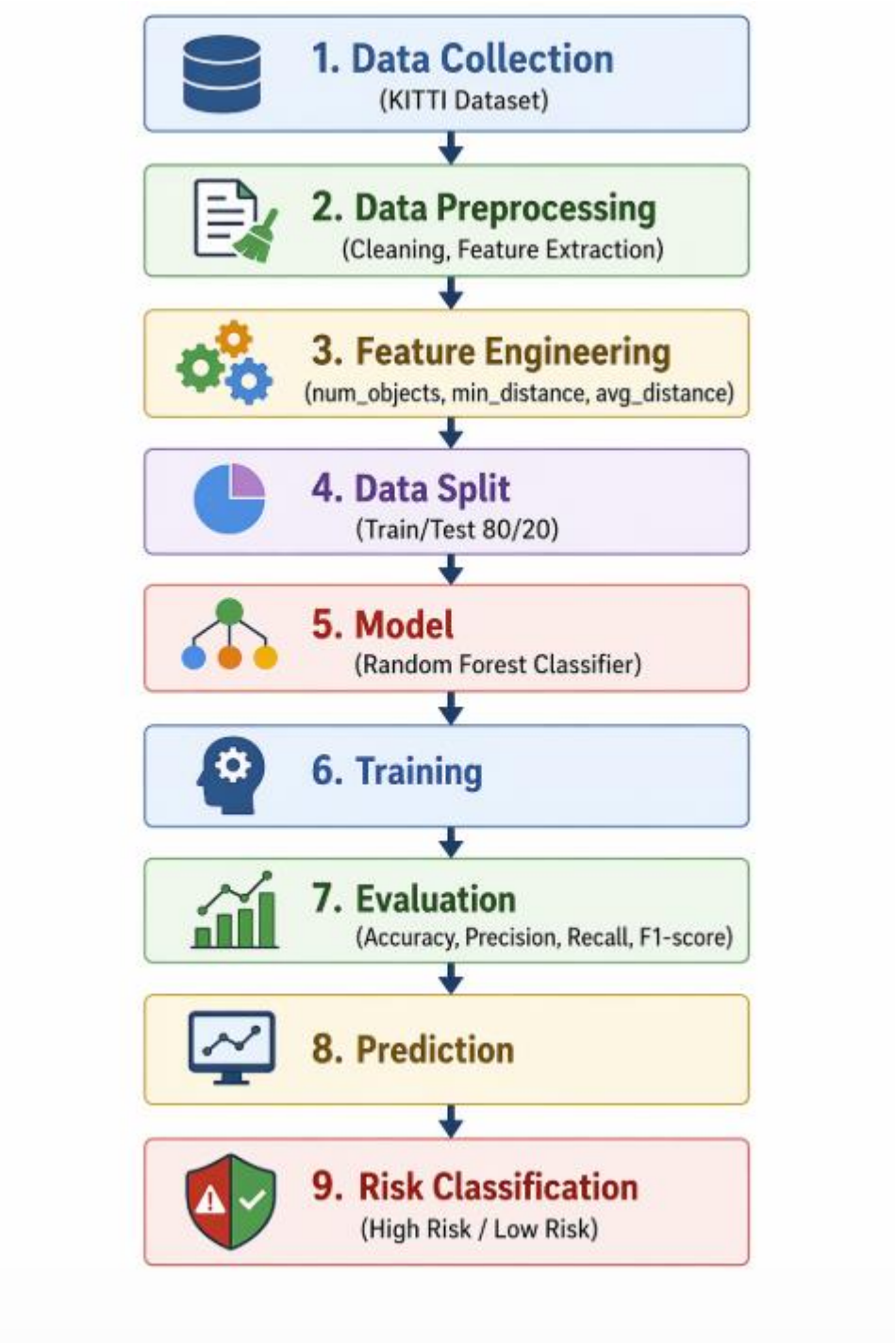
- Classifies scenarios as:
    - **High Risk (1)**
    - **Low Risk (0)**

Stage 5: Evaluation and Performance Metrics

The model is evaluated using standard classification metrics:

- **Accuracy:** Overall correctness of predictions.
- **Precision & Recall:** Performance in detecting high-risk cases.
- **F1-score:** Balance between precision and recall.
- **Confusion Matrix & ROC Curve:** Visualization of model.

2.5 Flow Chart



III – Dataset Selection

Public Dataset:

The dataset used in this project is the KITTI Vision Benchmark Suite (Object Detection Dataset), a widely used public dataset for autonomous driving research. Data wrangling was performed to transform raw label files into a structured format suitable for analysis. Relevant information was extracted from text files and converted into a tabular DataFrame, organizing key features such as object count and distance.

Data cleaning was carried out to ensure consistency and reliability of the dataset. Irrelevant entries labeled as “DontCare” were removed, and missing or empty values were handled by assigning default distance values. Additionally, the dataset size was limited to the first 500 samples to enable efficient processing. A target variable, collision risk, was created to convert the problem into a classification task, where high risk is assigned when object density is high or distance is very low, and low risk otherwise.

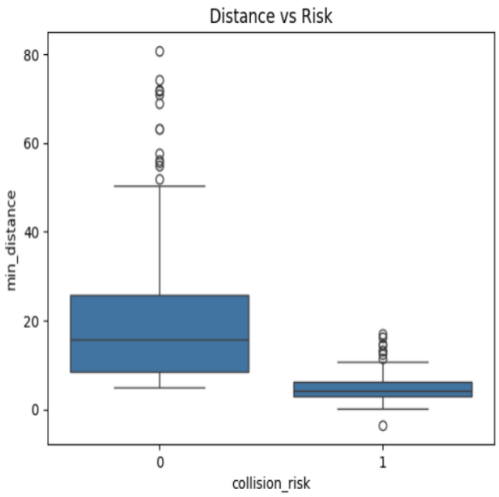
Normalization was applied to ensure that features are on comparable scales. Distance-based features are inherently normalized, and no significant outliers were observed that could impact model performance. Implicit normalization was achieved through careful feature selection, ensuring that the dataset is suitable for training and evaluation of the machine learning model.

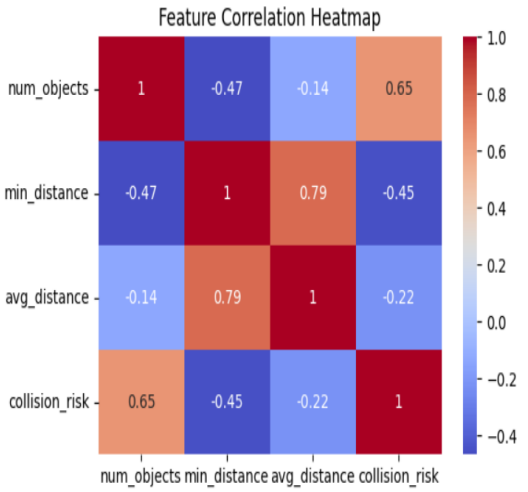
IV – Exploratory Data Analysis (EDA)

Exploratory Data Analysis (EDA) is performed to understand patterns in the dataset and how different features influence collision risk. Summary statistics show that num_objects represents traffic density, while min_distance and avg_distance indicate proximity to surrounding objects. Higher object density and lower distances are generally associated with high-risk scenarios.

Data visualization techniques help analyze feature relationships. Histograms display the distribution of object counts, boxplots compare minimum distance across risk categories, and heatmaps show correlations between features such as object count, distance, and collision risk. A confusion matrix is also used to evaluate model performance by comparing actual and predicted values.

Outlier detection identifies rare or extreme cases, such as very high object density or extremely low distances, which represent critical driving situations. These cases are limited but important for understanding high-risk conditions. Overall, EDA confirms that object density and distance are key factors in predicting collision risk.





V – Model Development

Among the available techniques, classification is selected as the most appropriate method for this project, as the objective is to predict discrete outcomes (high-risk or low-risk scenarios).

Classification

Classification is a machine learning technique used to categorize data into predefined classes or labels. It is widely used in autonomous vehicle systems for tasks such as risk assessment and decision-making.

In this project, classification is used to predict collision risk, where each scenario is categorized as:

- 0 → Low Risk
- 1 → High Risk

The model learns patterns from input features such as object density and distance to determine whether a situation is safe or dangerous. The Random Forest algorithm is used for this purpose, as it can handle non-linear relationships and provides accurate and reliable predictions for real-world data.

```
Sample 1: 0 → Low Risk
Sample 2: 1 → High Risk
Sample 3: 0 → Low Risk
Sample 4: 0 → Low Risk
Sample 5: 0 → Low Risk
Sample 6: 0 → Low Risk
Sample 7: 1 → High Risk
Sample 8: 0 → Low Risk
Sample 9: 0 → Low Risk
Sample 10: 0 → Low Risk
```

VI – Model Evaluation

The dataset is split into 80% training data and 20% testing data to evaluate model performance on unseen data. The model is assessed using metrics such as accuracy, precision, recall, F1-score, and support, which measure overall correctness and the ability to correctly identify high-risk scenarios. The support values show that the dataset contains 73 low-risk and 27 high-risk samples.

```
Accuracy: 1.0
              precision    recall  f1-score   support

           0       1.00      1.00      1.00        73
           1       1.00      1.00      1.00        27
```

Cross-validation and hyperparameter tuning are techniques used to improve model performance and generalization. Cross-validation validates the model on different data subsets, while hyperparameter tuning optimizes parameters like the number of trees in the Random Forest model.

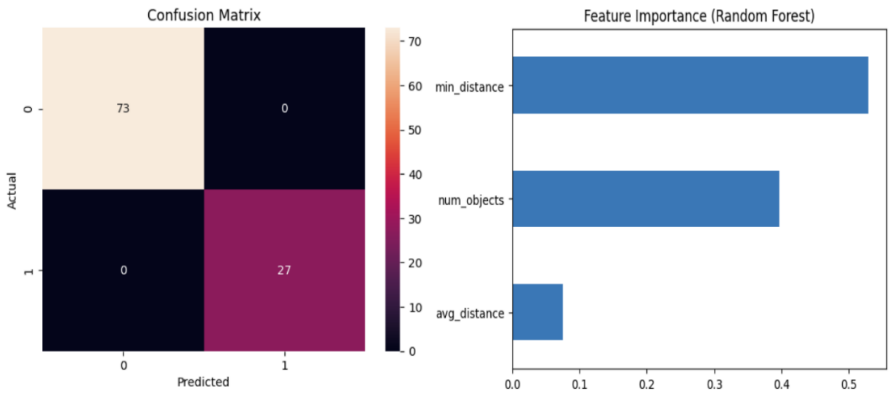
```
Cross-validation scores: [1. 1. 1. 1. 1.]
Average accuracy: 1.0
```

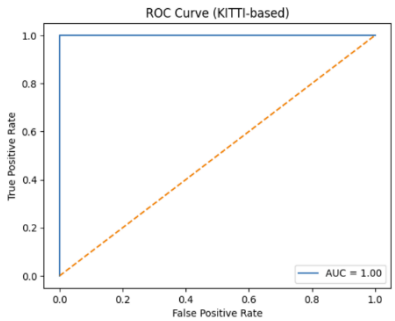
VII – Results & Insights

The model successfully predicts collision risk using features such as object density and distance. High-risk scenarios are identified when object count is high and the distance to nearby objects is low.

The results were visualized using a confusion matrix, ROC curve, and feature importance plot, which show that the model performs well in distinguishing between high-risk and low-risk scenarios. Distance was observed to be the most influential feature.

Compared to simple rule-based methods, the machine learning model provides better accuracy and improved prediction capability. While advanced models like YOLO and LSTM offer higher accuracy, they require higher computational resources. The current model achieves a good balance between performance, simplicity, and efficiency.





Limitations:

- **Simplified Risk Labelling:** Collision risk is generated using rule-based thresholds, which may not accurately reflect real-world uncertainty and variability in driving scenarios.
- **Lack of Temporal Analysis:** The model does not consider time-dependent relationships, meaning it cannot analyze how traffic situations evolve over time.
- **Absence of Sensor Fusion:** The approach does not integrate multi-sensor data (LiDAR, radar, camera), limiting the richness of environmental understanding.

Future work can focus on integrating real-time sensor and traffic data to improve prediction accuracy. Incorporating temporal (time-series) analysis will help capture dynamic changes in traffic conditions. The use of sensor fusion (LiDAR, radar, camera) can enhance environmental understanding.

VIII - Conclusion

This project successfully demonstrates the application of data analytics and machine learning techniques for predicting collision risk in autonomous ambulance systems. By utilizing the KITTI dataset, key features such as object density and distance were extracted and used to build a classification model capable of identifying high-risk and low-risk driving scenarios.

The Random Forest model showed effective performance in capturing patterns within the data and accurately classifying risk levels. The results highlight that lower distances and higher object density significantly contribute to increased collision risk, validating the importance of these features in safety-critical environments.

In conclusion, this underscores the importance of employing data analytics to make decisions and improve the level of safety in AVs.